FAIR 3D HERITAGE
Documentation, Publication and Preservation of the Digital 3D Heritage on the Web

How can MorphoSource, a Community 3D Repository, Support Educators and Museum Partners in Managing Student-Created Data?

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Abstract
While digital surrogates of museum objects can dramatically expand access to collections for diverse communities, creating and appropriately disseminating these surrogates can present significant practical challenges. This presentation critically evaluates a repository-based workflow developed for a class offered at Duke University, in which students participated in the creation and dissemination of digital surrogates of collection objects. This collaborative project generated both pedagogical value for the students and preserved digital surrogates for the museum.

The course "Virtual Museums: Theories and Methods of Twenty-First-Century Museums," offered in the autumn of 2025 at Duke University, introduced undergraduate and graduate students to museum exhibition and curation practices through the digitisation of objects from the Duke Nasher Museum of Art's collections. The students produced 3D models of African masks using photogrammetry and presented the models in a collaborative virtual exhibition within a 3D museum space that they designed. Each student was responsible for archiving the RAW source photographs and high-polygon mesh derivatives in MorphoSource, a community-oriented data repository headquartered at Duke University and specialised for 3D, 2D, and AV images representing physical objects of scholarly importance (MorphoSource 2026).

Through the duration of the course and after, MorphoSource's administrative tools supported shared access to and management of digital files by all project stakeholders. After creating a course project, the instructor assigned students as project members. As the students made their deposits, MorphoSource guided students through creating metadata and linked the records to the course project, represented museum objects, creators, and the Nasher Museum, while documenting relationships between source files and derivatives. By the end of the course, the student-created digital objects were

organised, described, and preserved within a repository environment. Museum staff could continue to access, track, and manage all deposits after the course ended.

In addition to supporting interactive viewing and examination, the workflow put the FAIR principles into practice (Wilkinson et al. 2016). Findability was supported through structured, modality-specific metadata, project organisation, institutional attribution, and repository indexing. Accessibility was managed through stable records and configurable preview and download permissions. Interoperability was supported through standardised metadata fields and persistent links among records. Reuse was strengthened by preserving source images alongside derivatives and documenting methods, software, creators, and relationships among files.

The workflow nevertheless revealed several limitations. Students required training and guidance to produce consistent metadata, and those unfamiliar with repository and archival practices found the submission process challenging. Compared with less structured file-sharing platforms, MorphoSource required greater investment during deposit, but provided stronger support for provenance, discovery, preservation, and continued institutional oversight. This case study therefore offers a practical model for using repository infrastructure to align pedagogical activities with the long-term stewardship of student-generated 3D cultural heritage data.

References

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